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From conduction to keyhole transition on copper using blue laser: Bead-on-plate process modeling and analysis of physical phenomena

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ABSTRACT

The growth of electric mobility has stimulated a large and expanding interest in achieving new manufacturing techniques that would enable more efficient production with less waste. Due to their high levels of flexibility, efficiency, and monitoring capabilities, laser-based processes are essential technologies in this context. Due to high laser absorption of copper for wavelengths under 450 nm, blue lasers are among the most promising laser sources currently on the market and are generating significant interest from battery manufacturers. From a manufacturing point of view, high absorption leads to the possibility of having conduction welding, with calmer melt pool, instead of keyhole welding, characterized by more turbulent molten pool dynamics and higher porosity. On the other hand, consolidated near-IR lasers only lead the possibility of keyhole welding due to poor copper absorptivity with 1064 nm wavelength. In this paper, the transition from conductive to keyhole regime was investigated and predicted through experimental and numerical analyses. From an experimental point of view, the transition is characterized by rapid change of laser absorption and the consequent variation of the seam geometry. Following experimental results, the surface vapor pressure threshold responsible for the transition to the keyhole regime is calculated in accordance with the process parameters employed. Finally, the comparison between experimental and predicted data confirmed the assumption that the recoil pressure is the main physical parameter determining the transition from conduction to keyhole mode, under certain conditions in terms of threshold fluence and irradiance.

1. Introduction

Laser welding of copper or copper-aluminum in dissimilar configuration have received much attention over recent years, in particular due to the high electrical conductivity of these materials useful in the area of e-mobility.

Within this field, extensively work has been done to optimize conventional joining process such as ultrasonic welding and friction stir welding. Good and reliable quality of Cu/Cu spot welded joint was obtained by Ni et al. (2022) who studied the joint formation mechanism and the major cause of joint failure. Same satisfactory results were presented by Kesharwani et al. (2022) for alluminiun/copper hybrid friction stir welding. Although it is possible to generate weld seams with good properties, these conventional technologies are characterized by low flexibility and repeatability. Within this context, a highly rated alternative for industrial application is autogenous laser welding, as clearly shown by Punzel et al. (2020) who compared different continuous-wave laser source for copper welding.

Despite being widely used in many different applications within the electronic and electrical industry, joining of copper presents several challenges due to the high reflectivity at common high-power industrial laser wavelength. Indeed, the absorption at room temperature of copper for near-infrared laser radiation is only 3–5% (Hügel and Dausinger, 2004). This value increases with copper surface's temperature increasing (Hügel and Dausinger, 2004) and with the phase transition from solid to liquid but at the beginning of the welding process the required power is very high. Furthermore, the high thermal conductivity of copper causes the dissipation of the heat input and as a result a difficult in-depth penetration of the welding. This second feature confirms that a significant heat input or preheating process is required to achieve a sufficiently deep penetration depth. Laser power of near-infrared laser sources has improved their quality significantly over the last decade, and today's high-power 10 kW source can be concentrated on a spot of 100 μm, resulting in high power densities. High-brightness laser sources have improved the ultimate performance of copper welding in terms of penetration depth, but spatter, melt

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The amount of energy delivered to the target was calculated using the target material's absorption coefficient. In particular, in accordance with the literature, copper's absorption with the interaction of the blue wavelength was initially thought to be equal to 0.6 and was later modified to obtain the best prediction of the geometry of the molten and resolidified material in terms of width and depth.

The heat capacity and thermal conductivity of solid copper were determined experimentally up to 300 °C using a comparative method (ASTM E 1461–07) and LFA 447 Nano Flash Netzsch equipment. Based on literature considerations, the same properties were defined after 300 °C, as well as the density and the coefficient of thermal expansion. Table 2 reports in detail all the temperature-dependent properties.

All these physical parameters are then utilized in Fourier's equation (Eq. 2) to resolve the time-dependent temperature distribution in the material during laser interaction.

$$\rho C_p \frac{\partial T}{\partial t} + Q = \nabla(k \nabla T) \tag{2}$$

Where ρ is the copper density [kg/m^3], Cp is the heat capacity at constant pressure [J/(kg· K)], k is the thermal conductivity [W/(m· K)], Q the absorbed energy density [W/m^3], T the temperature [K] and t the time [s]. Radiant heat losses were considered negligible, while convective heat losses in air were considered setting an ambient air temperature of 25 °C.

As concern the mushy zone, where solid and liquid phases coexist, the material properties were redefined by the software on the basis of solidus and liquidus temperature following Eqs. (3)–(5) where θ_s and θ_l are the phase fraction of solid and liquid material respectively, α_m is the mass fraction and $L_{s\rightarrow l}$ is the latent heat absorbed from solid phase to liquid phase (207 kJ/kg).

$$\rho = \theta_s \rho_s + \theta_l \rho_l \tag{3}$$

$$C_p = \frac{1}{\rho}\left(\theta_s \rho_s C_{p,s} + \theta_l \rho_l C_{p,l}\right) + L_{s\rightarrow l}\frac{\partial \alpha_m}{\partial T} \tag{4}$$

$$k = \theta_s k_s + \theta_l k_l \tag{5}$$

Tetrahedral mesh elements are used in the whole domain. Since the physic phenomenon principally involves the area along the laser path, this domain is meshed with finer size elements than the surrounding part. Far from the welding area, elements with maximum dimension of 0.5 mm were set, while a mesh dimension optimization was performed close to the laser path to gain enough confidence in the accuracy of the model while not exceeding the available computational resources. A size-controlled parameter was set inside volumetric block domain with dimension 1×1x 2.3 mm^3 defined in the middle of the laser path Fig. 2b. The parameter "m" defines the maximum element size inside this volume and a parametric sweep study was set changing m from 40 to 200 μm. From this analysis a maximum element size was set equal to 80 μm, since lower values had negligible effect on the final results, in terms of melt pool dimension (<2%), as show in Fig. 2c.

The influence of liquid flow on the temperature field significantly affects the dimension of the phase-change volume thus, in a second time, a refinement of the thermal model was performed by adding the effect of fluid motion inside the weld pool. The Computational Fluid Dynamics (CFD) model was set up by activating the natural convection and Marangoni effect while, the effect of vapor loss on thermal profile was not considered. The viscosity data is further required in CFD model and the input data are reported in Table 2.

3. Results

3.1. Experimental analysis

3.1.1. Geometry

Representative cross sections of the laser processed path obtained with different process parameters (Power, Laser speed, Irradiance and Fluence) are shown in Fig. 3, while mean measurements of depth and width, respectively in column 5 and 6, of the melted area are reported in Table 3. As expected, the interaction area dimension decreases as the laser moving speed increases since the heat input, in terms of linear energy density, decreases as well. Maximum depth (d) of 0.78 mm was measured with laser speed of 5 mm/s and power of 1500 W and the correspondent width (w) was 0.82 mm, with a resulting aspect ratio (d/w) of 0.95.

After a more accurate analysis of the geometric data, it is possible to

Table 2
Copper temperature-dependent thermophysical properties at solid and molten state (Baba et al. (2012), Assaela et al. (2010)). Grey cell contains the results of experimental data performed in this work.

T [K]	Density [kg/m^3]	Viscosity [mPa•s]	Surface tension	Heat Capacity [J/kg•K]	Thermal conductivity [W/m•K]
300	8930	/	/	407.0	330.6
350				409.0	323.6
400				410.0	316.7
450				411.0	310.0
500				412.0	303.5
550				413.0	297.1
600				414.0	290.9
700				416.6	278.8
800				419.0	267.4
900				421.2	256.5
1000				423.5	246.3
1100				425.8	236.7
1200				428.1	227.6
1300				430.4	219.2
1356 (solid)				431.7	214.7
1356 (liquid)	7998	4.03	1313.2	3300.0	157.2
1400	7962	3.74	1305.6	800.0	159.4
1500	7881	3.21	1288		164.4
1600	7799	2.81	1270.4		169.3
1700	7717	2.50	1252.8		174.3
1800	7635	2.25	1235.2		179.3
1900	7553	2.05	1217.6		184.3
1950	7512	1.96	1206		186.7
2000			1200		
3000			1024		
3500			936		

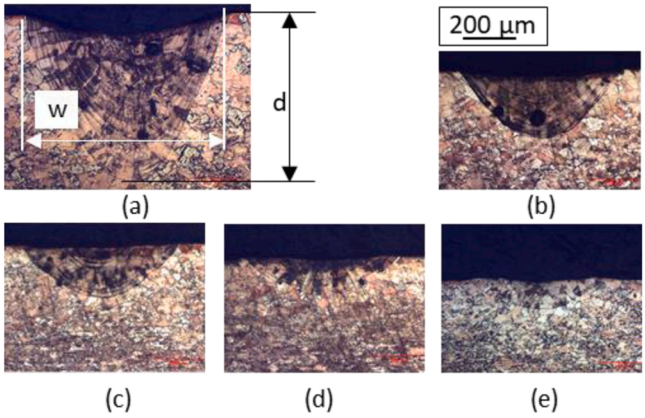


Fig. 3. Geometry of the laser-copper interaction area of 1350_5 (a), 1350_10 (b), 1350_30 (c), 1350_50 (d), 1350_70 (e) samples.

Table 3
Summary of the average measurements of depth and width as the process parameters vary (P = Power, v=laser speed, I=Irradiance, F=Fluence, d=seam depth, w=seam width).

1 P [W]	2 v [mm/s]	3 I [MW/cm^2]	4 F (x10^4) [J/cm^2]	5 d [mm]	6 w [mm]	7 Aspect Ratio (d/w)
1500	5	0.80	7.80	0.78	0.82	0.95
1500	10	0.80	3.90	0.57	0.77	0.74
1500	30	0.80	1.30	0.21	0.61	0.34
1500	50	0.80	0.78	0.16	0.59	0.27
1500	70	0.80	0.56	0.14	0.57	0.25
1350	5	0.72	7.02	0.64	0.79	0.81
1350	10	0.72	3.52	0.33	0.64	0.52
1350	30	0.72	1.17	0.2	0.59	0.34
1350	50	0.72	0.70	0.14	0.56	0.25
1350	70	0.72	0.50	0.13	0.55	0.24
1200	5	0.64	6.24	0.43	0.66	0.65
1200	10	0.64	3.12	0.24	0.6	0.4
1200	30	0.64	1.04	0.15	0.52	0.29
1200	50	0.64	0.62	0.11	0.47	0.23
1200	70	0.64	0.45	0.1	0.45	0.22

highlight some interesting aspects. In Table 4 the degree of association between geometrical and process parameters is measured by a correlation coefficient, denoted by R (or Pearson's correlation coefficient). The correlation coefficient is measured on a scale that varies from + 1 through 0 to − 1. Complete correlation between two variables is expressed by either + 1 (positive correlation) or − 1 (negative correlation), while complete absence of correlation is represented by 0. In Table 4 five classes of correlation were highlighted from very weak (light blue) to very strong (dark blue) correlation. The results show a very strong correlation of the seam geometry, in particular its depth, with laser fluence, while power (or irradiance) have a marginal effect on the final depth (weak) and a moderate impact on the width.

3.1.2. Defects

Defect analysis and in particular porosity percentage inside the melted and resolidified copper was correlated to the fluence. By classifying the fluence values used in three classes as follows: 1–2.5 (x10^4) J/cm^2, 2.6–5 (x10^4) J/cm^2, > 5 (x10^4) J/cm^2 it is evident that the porosity increases when the fluence increases with the following measured mean values: 0.7%, 2.6%, 3.5%. In all cases, the percentages of pores are below 5%, however several large pores have been detected in some sections of the specimens, mainly if higher fluences were defined, in agreement with the average values presented. An example of these macro pores is shown in Fig. 4. These defects can be categorized as "keyhole stability defects" in accordance with the increased porosities with fluence and the identified geometry of these pores, which appears rounded in shape. In fact, the stability of the keyhole and the ultimate weld-bead quality are strictly correlated in deep penetration welding of metals. The keyhole becomes unstable and collapses as the heat input rises (greater fluence), isolating the gas inside and causing bubbles to form and resulting spherical porosities.

Analyzing the micrograph sections of welded beads, it was possible to highlight that the fluence does not only affect the geometry of the bead and the percentage of porosity, but also the position of the pores. In order to evaluate a correlation between laser parameters and porosity distribution, the bead was virtually divided into two parts: a central area (C) with depth and width equal to 80% of the bead size and an interface area (I) as shown in Fig. 5 as an example. For both areas, the relative porosity was calculated and this data was correlated with the fluence values for each sample.

This correlation shows an increasing of porosities in the interface between base-material and melted material at fluence decreasing, while for high fluences most of the pores are concentrated within the bead. This relationship is shown in Fig. 6, and the relative porosity, which is determined by dividing the porosity in the area of interest by the total porosity of the joint, increases with fluence in the central area (it will complementarily decrease in the area at the interface).

3.1.3. Microstructure and hardness

Samples 1500_5 and 1500_50 were used in this section in order to show the weld beads properties in terms of microstructure evolution and hardness. The weld bead grain morphology in the fused zone (FZ), in the heat affected zone (HAZ), and in close proximity to the base material are highlighted in Fig. 7 and Fig. 8 by optical micrographs and SEM magnification. Equiaxed grains at the FZ's core and close to the surface are followed by highly elongated grains oriented in the direction of the heat flow and which extend up to the HAZ. In the HAZ coarser elliptical-shaped grains and abundant annealing twins are visible and their dimension decrease moving away from the joint. The chemical composition of the base material and inside the weld was carried out by EDS analysis and the results (Table in Fig. 7) confirm the purity of the copper sheet.

According to the results of the microstructural analysis, the hardness measurements on the 1500_5 exhibit higher hardness (mean 114 HV0.1,

Table 4
Correlation factor between geometric and process parameters.

	Depth (d)	Width (w)	d/w
Power/Irradiance	0.32	0.51	0.26
Speed	-0.77	-0.76	-0.82
Fluence	0.94	0.87	0.96

Very strong
Strong
Moderate
Weak
Very Weak

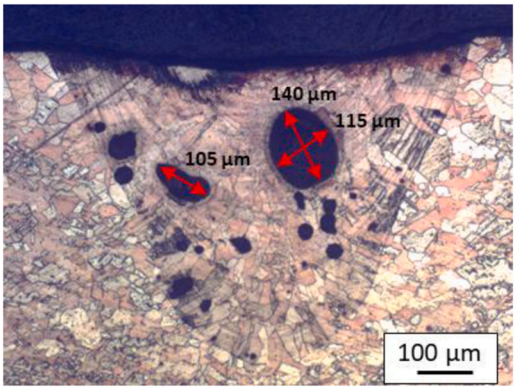


Fig. 4. Representative cross-section (1500_10) for the evaluation of large pores dimension.

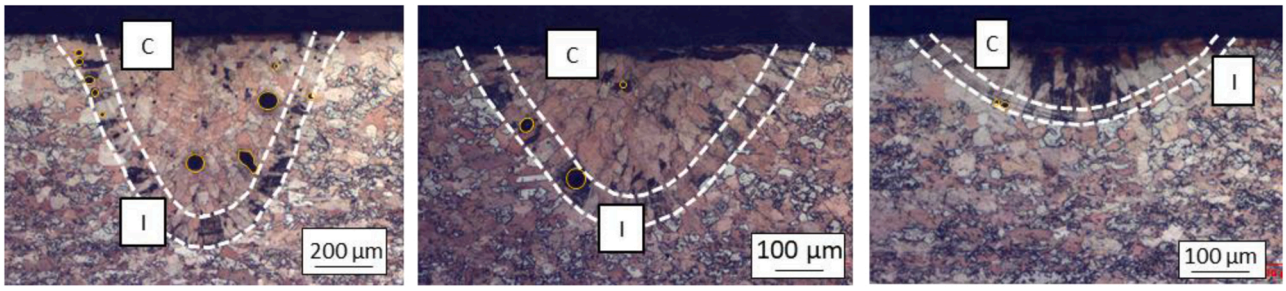


Fig. 5. Internal-external porosity measurement method. From the left the cross section of 1500_5, 1500_30 and 1500_50 samples are shown.

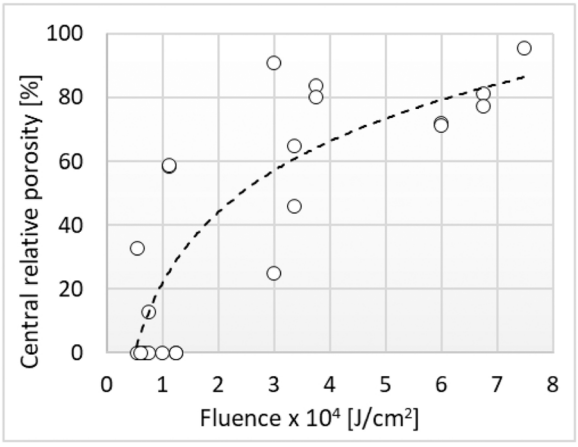


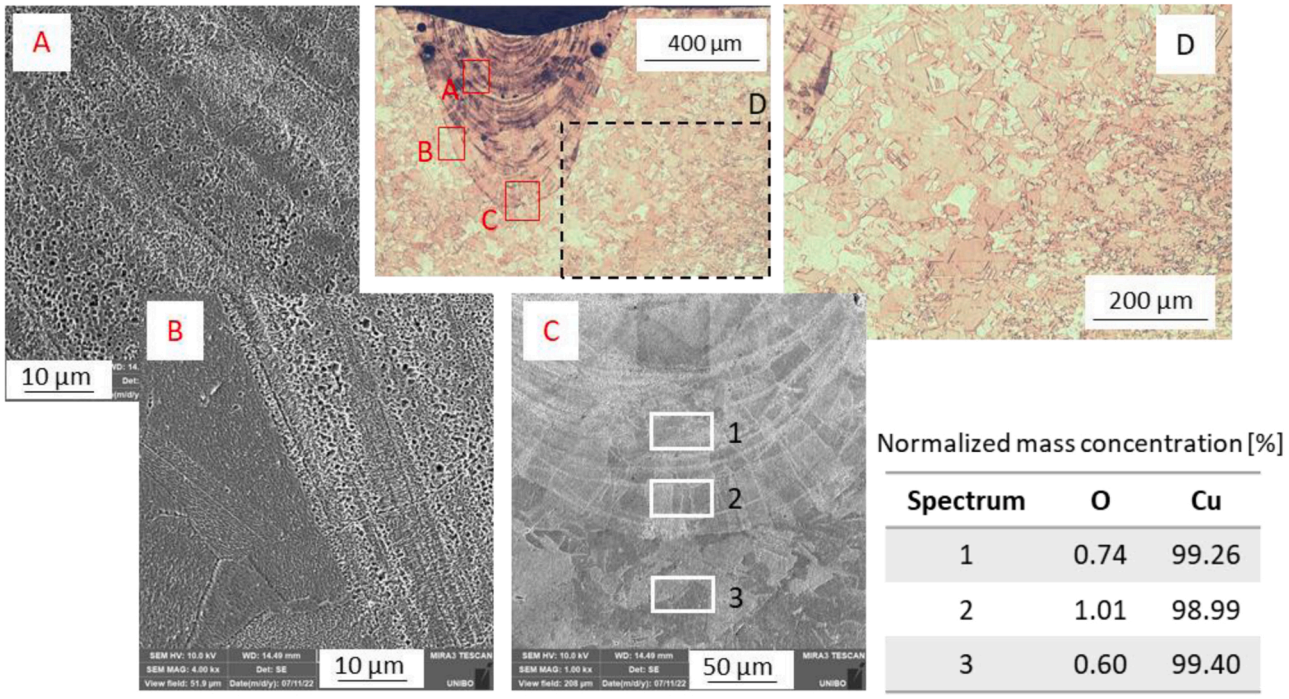
Fig. 6. Correlation between relative porosity in the central (C) and interface (I) areas for different fluence levels.

standard deviation 5.8) in the melt pool, lower values (mean 67 HV0.1, standard deviation 3) in the HAZ, and intermediate values (mean 78 HV0.1, standard deviation 2.8) in the base material. Hardness measurement of 1500_50 sample confirm the same match with microstructural investigation and following values were identified: 98 HV0.1 in the melt pool, 70 HV0.1 in the HAZ and 76 HV0.1 (standard deviation 2.4) in the base material. The standard deviation cannot be calculated in that case due to the small size of the melt pool and HAZ. Fig. 9 displays the longitudinal and transverse profiles of hardness of both samples. The extension of the HAZ may be determined from the hardness profile and, for maximum fluence, reaches widths of 2.7 mm near the surface and 1.3 mm in depth, while for reduced fluence, the corresponding values are 0.5 mm and 0.8 mm.

3.2. Model analysis

As was already mentioned, a parametric simulation was first run, and the absorption coefficient was varied to range from 0.20 to 0.60 for each set of process parameters. Columns 5 and 6 in Table 5 contain the results of the laser-material interaction's area in terms of predicted depth (d_M) and width (w_M), comparable with experimental measurements. The simulation that best predicts the real geometry has been chosen, and the relative absorption coefficient value is reported in column 8. Instead, Column 7 displays the average percentage error calculated on the width and depth of the area of interest, expressed as an absolute value.

Plotting the data of the resulting absorption coefficient and the calculated prediction error with respect to the fluence used for each test



Normalized mass concentration [%]

Spectrum	O	Cu
1	0.74	99.26
2	1.01	98.99
3	0.60	99.40

Fig. 7. 1500_5 microstructure inside the melted and resolidified area (A-B), in the interface edge (C) and in the heat affected area (D) captured by OM and SEM. Microanalysis in three different small area of the samples is also shown in the table.

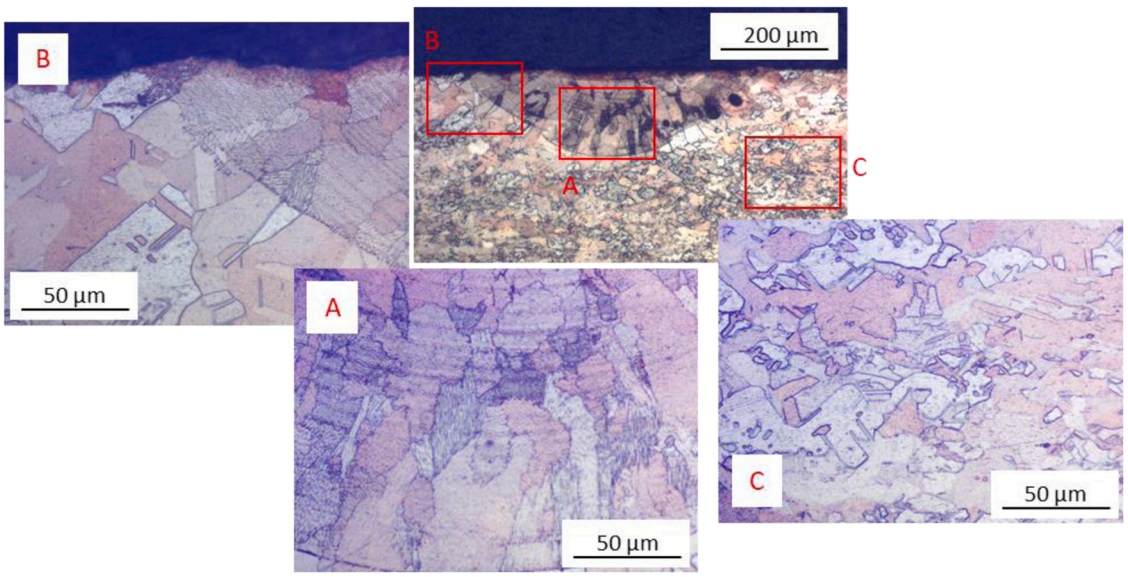


Fig. 8. 1500_50 microstructure inside the melted and resolidified area (A), in the interface edge (A) and in the heat affected area (C).

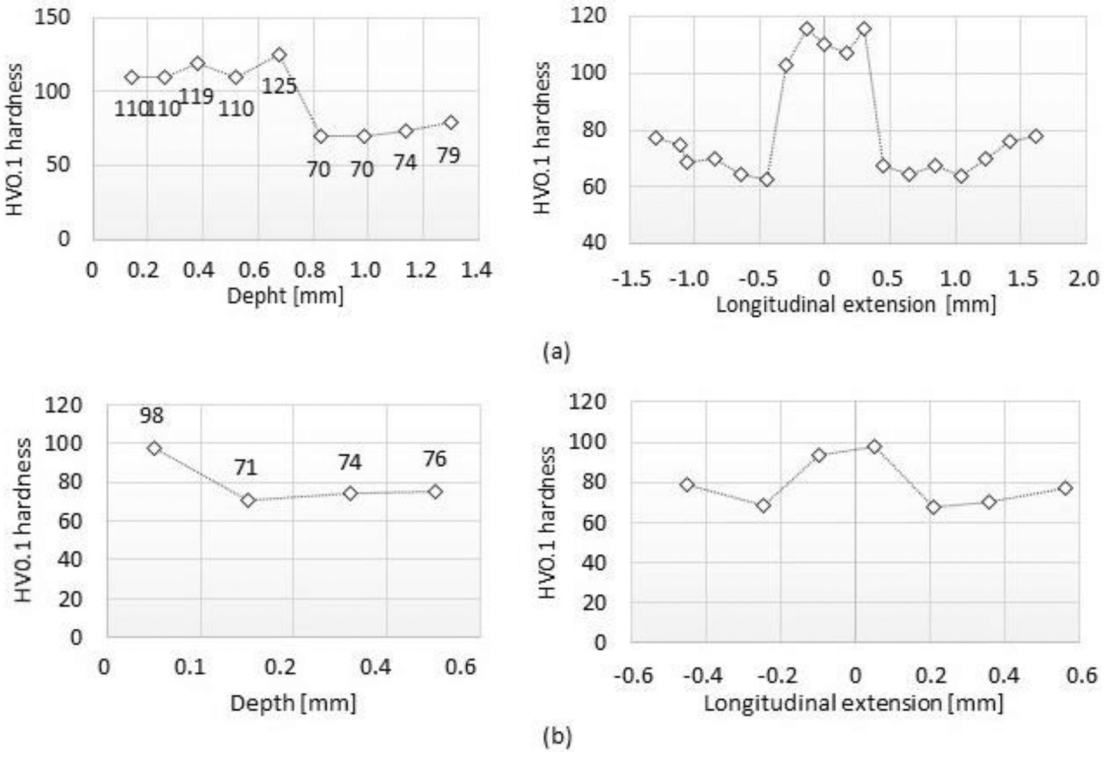


Fig. 9. Longitudinal and transverse hardness profile carried out on 1500_5 (a) and 1500_50 (b) respecting the scheme shown in Fig. 1.

both these factors increase linearly. This relationship is clearly shown on the graph in Fig. 10, and the regression line that was obtained has a very high correlation factor, greater than 0.8 for both variables.

From Fig. 10 analysis is possible to highlighted two different issues: (1) the obtained absorption coefficients (from 0.23 to 0.35) are significantly lower than expected values (around 0.6) and (2) average geometric error predicted by the models significantly increases with fluence up to values greater than 30%.

As previously stated, the prediction error has been provided as the average of the error between width and depth expressed as an absolute value. However, to better understand what causes the increasing of the error, an analysis of the aspect ratio (depth/width) of the real and simulated melted zones is required, and Fig. 11 permits a more intuitive comparison. The figure depicts a simplified geometry of the real (green) and simulated (red) melted zones in the form of a table. All boxes with a difference of more than 15% between these two curves have a gray cross at the bottom left highlighted. In all these cases, the real fused areas are deeper and narrower than the simulated ones, resulting in higher aspect ratios. The differences in the aspect ratio between real measurement and calculated simulation increases with fluence because the predicted melted area is less sensitive to laser process parameters as shown in Fig. 12 based on the comparison between column 7 of Table 3 and

Table 5
Summary of depth and width calculated by FE simulations (columns 5–6), comparison with experimental data (column 7), resulting absorpion coefficient (column 8) and aspect ratio (column 9).

1 Test	3 I [MW/cm^2]	4 F x10^4 [J/cm^2]	5 d_M [mm]	6 w_M [mm]	7 |e| [%]	8 Ca	9 Aspect Ratio $(d/w)_M$
1500_5	0.80	7.80	0.49	1.09	35.4	0.33	0.45
1500_10	0.80	3.90	0.42	0.98	26.8	0.35	0.43
1500_30	0.80	1.30	0.20	0.62	3.5	0.25	0.32
1500_50	0.80	0.78	0.16	0.59	0.3	0.24	0.27
1500_70	0.80	0.56	0.14	0.55	2.5	0.23	0.25
1350_5	0.72	7.02	0.45	1.04	30.2	0.35	0.44
1350_10	0.72	3.51	0.28	0.75	15.9	0.3	0.38
1350_30	0.72	1.17	0.19	0.61	4.8	0.27	0.31
1350_50	0.72	0.70	0.13	0.54	3.9	0.24	0.25
1350_70	0.72	0.50	0.12	0.52	5.8	0.24	0.23
1200_5	0.64	6.24	0.31	0.81	24.9	0.32	0.39
1200_10	0.64	3.12	0.22	0.66	10.0	0.29	0.33
1200_30	0.64	1.04	0.14	0.55	6.4	0.26	0.24
1200_50	0.64	0.62	0.11	0.50	3.9	0.25	0.22
1200_70	0.64	0.45	0.10	0.48	3.4	0.25	0.21

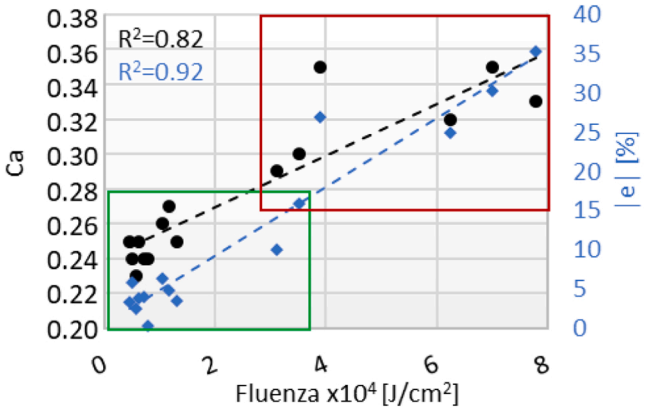


Fig. 10. Absorption coefficient resulting from the comparison between real and simulated data at the variation of fluence and relative average error.

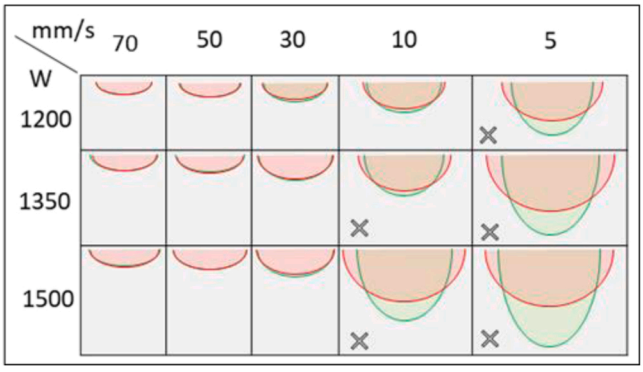


Fig. 11. Melt pool geometry at different laser power and laser speed: measured (green) and simulated (red) dimensions.

column 9 of Table 5.

A further analysis of the resulting data was necessary to ensure that the discrepancy between model and reality was not due to an over-simplification of the model. The effect of CFD physics on absorption coefficient calibration and melted area aspect ratio was evaluated and for comparison the 1350_5 model was used. Non-isothermal and heat transfer modulus have been combined to simulate fluid flows with

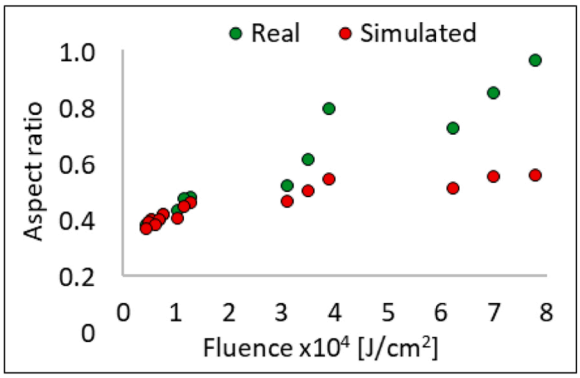


Fig. 12. Aspect ratio (depth/width) of the melt pool geometry: measured (green) and simulated (red).

temperature-dependent fluid properties. Fig. 13a depicts this effect, with a significant decrease in width (−26%) and depth (−21.8%) of the melted area clearly visible. The maximum temperature falls from 3680 °C to 2662 °C. To restore the actual geometry prediction, the absorption coefficient was increased from 0.35 to 0.46 (+31%) and the temperature field in a mid-section of the samples is shown in Fig. 13b.

The definition of a more complex model that takes into account the dynamics of the melt pool and, in particular, the Marangoni effect and buoyancy forces, has a significant influence on the dimension of the melt pool if the model is not further calibrated, but not on the aspect ratio. The calculated aspect ratio continues to be 0.44 as in the thermal model and far from the measured aspect ratio (0.81).

4. Discussion

The results of the proposed model, when compared to those obtained experimentally, show that the model can correctly predict (error < 10%) the size of the volume of molten and resolidified material only for lower fluence values; specifically, fluences less than 3.5×10^4 J/cm^2. This discrepancy was not resolved by including a temperature-dependent laminar flow effect in the model. In fact, the introduction of CFD physics enabled to identify absorption coefficient values that were closest to those expected (which being between 0.6 and 0.7), but it did not provide a solution to the average error on the size. In the case 1350_5, used as an example, the aspect ratio of the melted zone is equal to 0.46, with a minimum variation compared to the model without laminar flow (0.44) and thus far from the true value of 0.81. The solution to this discrepancy between model and real interaction must thus be found in the different mode of interaction, which moves from conduction to keyhole. This is demonstrated analytically by determining whether the energetic conditions are actually attributable to an initial phase of keyhole triggering. The following threshold parameters can be determined from the literature and are required for the transition to keyhole:

- Irradiance: Cunningham et al. (2019) stated that a material-dependent critical irradiance on the order of 0.5 MW/cm^2 can be defined, over which the keyhole mode is active.
- Aspect ratio: Cunningham et al. (2019) and Nakamura et al. (2000) stated that keyhole mode interaction occurs when the aspect ratio is higher than 0.5.
- Exceeding of the boiling temperature for enabling the vapor surface pressure to cause a keyhole formation as mentioned by Wu et al. (2023): in case of pure copper the boiling temperature is 2562 °C.

According to Table 3, all the tests were performed with irradiance values greater than 0.5 MW/cm^2, implying that the energy density supplied to the laser-material system is sufficient to generate the keyhole

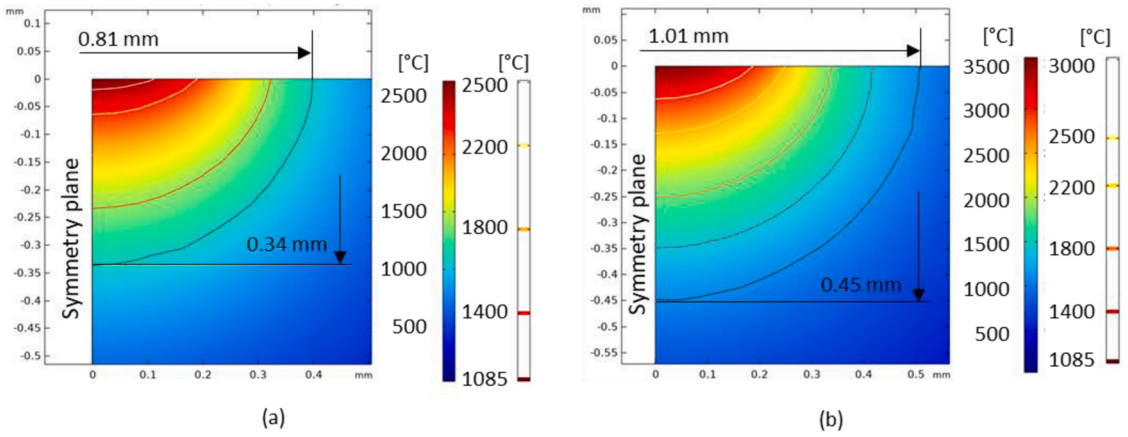


Fig. 13. CFD main effects impact on melted area geometry and aspect ratio before (a) and after (b) absorption coefficient re-calibration.

in all conditions tested. In contrast, aspect ratios value greater than 0.5 were only measured in some samples, and these correspond exactly to the tests for which there is a divergence between model and experiments. For these tests the maximum superficial temperature and the temperature heating rate, after melting, were calculated and resulting values are reported in Table 6.

As stated, confirms that all the necessary requirements were verified in 1500_5, 1500_10,1350_5,1350_10 and 1200_5 process conditions.

In order to univocally settles the conduction-to-keyhole transition, the vapor surface pression must be assessed. In fact, following Cunningham et al. (2019), once the surface temperature reaches the boiling point, localized and superficial vaporization formed a depression, which will in turn originate in the keyhole. The surface pressure can be calculated with Clasius-Clapeyron equation (Eq. 6):

$$p_s = p_0 e^{\Delta Hm/k_B(1/T_b - 1/T_s)} \tag{6}$$

Where p_0 is the working ambient pressure and the pressure at which the boiling point is defined (1 atm); ΔH is the heat of vaporization per atom (5.069×10^6 [J/kg]); m is the copper atomic mass ($1{,}0552061 \times 10^{-25}$ [kg]); k_B is the Boltzmann's constant ($1{,}380649 \times 10^{-23}$ [J/K]); T_B is the boiling temperature (2835 [K]) and T_s is the surface temperature.

Clasius-Clapeyron allows calculating the surface pressure with the assumption of constancy of the material volume due to the high heating rate ($>10^4$ K/s); heating rates in the order of magnitude of those actually calculated for the conditions under study (Table 6).

From the models' results it is possible to calculate the vapor surface pressure (p_s) considering the maximum reached temperature as shown in Fig. 14.

The pressure must be greater than the ambient pressure for the vapor depression to penetrate the melt pool and causes a significant displacement of liquid from the laser-material interaction region. Based on the results, the 1200_5 and 1350_10 tests represent a near-transition condition, whereas the pressure of 22.8 bar reached by the 1350_5 parameters can be considered the first case of this study in which a keyhole regime is completely reached. Consequently, also fluences' threshold for conduction-to-keyhole transition and keyhole obtainment can be calculated and their values are respectively 3.52×10^4 J/cm^2 and 7.02×10^4 J/cm^2.

The results obtained from the model's data analysis also allow for some additional observations on the microstructure analysis and the

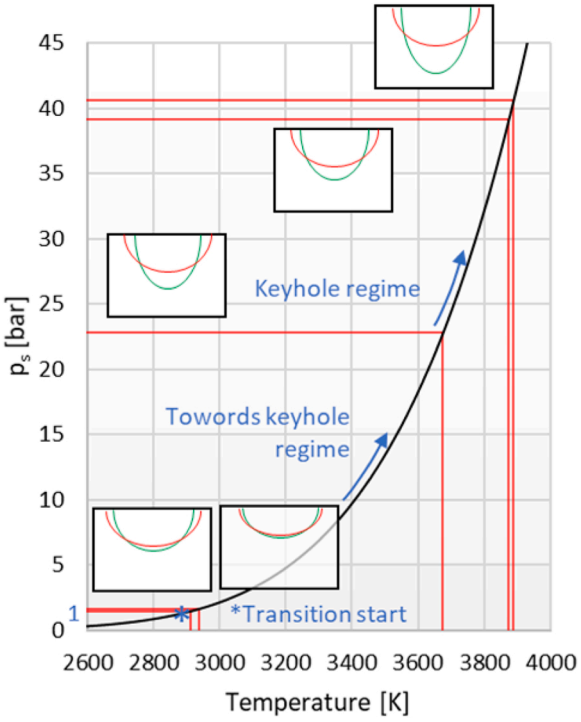


Fig. 14. Surface pressure at temperature variations calculated using Clasius-Clapeyron equation and p_s values calculated for the five cases of interest.

defects identified in the components following the interaction with the laser. Indeed, at lower fluence, the high thermal conductivity of copper allows a rapid heat transfer, preventing the large concentration of energy in the melted volume and limit the formation of pores due to metal vapor formation. However, the melted metal solidifies more quickly at the interface with the base material and if there was gas inside the melt pool it doesn't have the time to be expulsed from the weld, remaining trapped in the external part of the solidified metal. At fluence increasing the re-solidification velocity is lower while turbulence effects, Marangoni convection processes, buoyancy forces and surface boiling may

Table 6
Calculated maximum temperature and heating speed of divergence models.

	1500_5	1500_10	1350_5	1350_10	1200_5
Max Temperature [°C]	3889	3875	3676	2912	2939
Heating rate [°C/s]	6.19×10^4	2.76×10^4	2.72×10^4	4.48×10^4	2.28×10^4

disperse pores in different directions along the translation of the laser beam. Because of all of these factors, the presence of pores within the melted area is highly probable.

5. Conclusions

The effects of high-power blue laser source interaction in 2.3 mm thick commercial pure copper sheets were investigated to identify the conduction-to-keyhole regime transition. Experiment-based tests and simplified numerical simulation were used to identify the change of the interaction mode by evaluating the geometry of the laser-affected volume.

The comparison of experimental and numerical results revealed a very precise fitting of the data until the interaction regime is of complete conduction, allowing for the identification of a linear relationship between the absorption coefficient and the laser fluence. As soon as the laser-material interaction achieve the transition regime, analyzed data lose their convergence, and the error (in terms of molten volume) grows linearly with the increase in the energy density supplied to the system. The non-convergence of the data, because of the transition from conduction-to-keyhole regime, was verified starting from the analysis of three process conditions deemed necessary for the occurrence of the keyhole itself:

- Critical irradiance on the order of 0.5 MW/cm^2 is used for all test conditions.
- Divergence tests are characterized by melted and resolidified sections with aspect ratios greater than 0.5.
- Under the same process conditions of the preceding points, the boiling temperature has always been outdated.

Starting from these statements vapor surface pressure was calculated using Clasius-Clapeyron and a clear transition from conduction to keyhole mode was revealed when the calculated surface pressure reaches 22 bar. Based on these findings, it is possible to conclude that using current blue laser sources (BPP 30 mm/mrad and maximum power 1500 W), it is possible to identify process conditions in the keyhole regime during the interaction with pure copper if surface vapor pressure exceeds 22 bar and the energy density assumes value in the order of 7×10^4 J/cm^2.

CRediT authorship contribution statement

Erica Liverani: Conceptualization, Methodology, Writing – original draft. **Alessandro Ascari**: Investigation. **Luca Tomesani**: Supervision. **Alessandro Fortunato**: Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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